

test_02_constrained.vr

\$VCS_HOME/doc/examples/nativetestbench/openvera/rvm/host_if/tests/test_02_constrained.vr

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/******
*
* Author:      Fabian Delguste
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*
* This test shows how to create constrained test by defining
* a set of scenarios
*
* The scenario implemented here performs the following
* transactions:
*   - Generates 32 WRITE cycles at address [0..31]
*     with data 0x00, 0xFF, 0x55 or 0xAA
*   - Generate N WRITE cycles at the read-only addresses [0..7]
*   - Generates 32 READ cycles at address [0..31]
*
*****
*/

#include "dut_env.vrh"

// Define a constrained scenario
class constrained_intel_mgmt_scenario extends intel_mgmt_scenario {
    static bit [31:0] WRITES    = 1;
    static bit [31:0] WRITE_RO = 2;
    static bit [31:0] READS     = 3;
    constraint valid_scenarios {
        kind <= READS;
    }
    constraint writes {
        if (kind == WRITES) {
            length in {1:8};
            foreach (items, i) {
                items[i].kind == intel_mgmt_trans::WRITE;
                items[i].data in {8'h00, 8'hFF, 8'hAA, 8'h55};
            }
        }
    }
}
```

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    }
}
constraint write_ro {
    if (kind == WRITE_RO) {
        length in {1:4};
        foreach (items, i) {
            items[i].kind == intel_mgmt_trans::WRITE;
            items[i].addr in {0:7};
        }
    }
}
constraint reads {
    if (kind == READS) {
        length in {1:8};
        foreach (items, i) {
            items[i].kind == intel_mgmt_trans::READ;
        }
    }
}
task pre_randomize()
{
    super.pre_randomize();

    // Pre-allocate enough scenario items
    // for the maximum-length scenario
    this.resize_items(8);
}
}

program constrained {
    // Top level environment
    env the_env = new;
    // Insert the new scenario descriptor
    the_env.build();
    {
        constrained_intel_mgmt_scenario constrained_sc = new;
        the_env.gen.randomized_sc = constrained_sc;
    }

    // Kick off the test
    the_env.run_t();
}

```